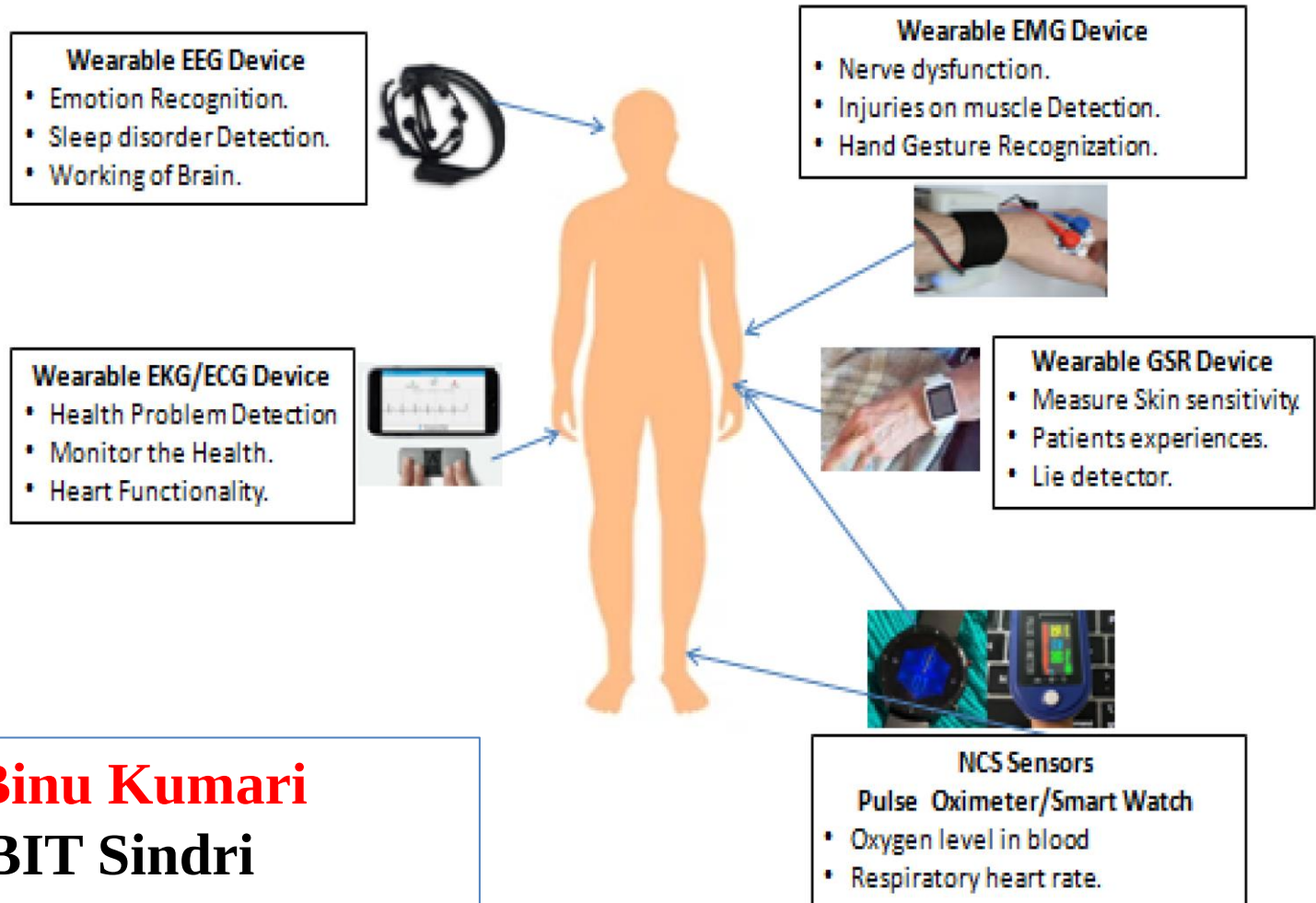


Bio signals



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Biomedical SIGNAL PROCESSING

- Application of engineering principles and techniques to the medical field to close the gap between engineering and medicine.
- Guide the medicine to use innovative educational tools such as humanistic models, realistic simulations, web-based online resources, etc.
- It combines the design and problem solving skills of engineering with medical and biological sciences to improve healthcare diagnosis and treatment.



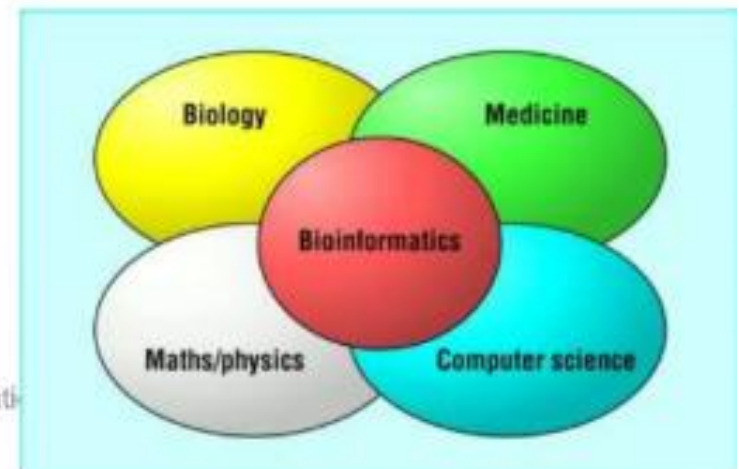
The Need...

- Medical decision-making requires the clinician to apply accumulated knowledge to a specific amount of patient information to produce a result that may be
 - a diagnosis,
 - prognosis,
 - course of therapy, or
 - the selection of further tests.
- Limited resources - increased demand
- Insufficient time available for diagnosis and treatment.
- Need for systems that can improve health care processes and their outcomes in this scenario



Sub-disciplines

- Bioinstrumentation
- Biomaterials
- Biomechanics
- BioInformatics
- Biomedical computing & signal processing
- Medical Imaging
- Orthopaedic Bioengineering
- Rehabilitation Engineering
- Minimally invasive surgery



Biosignals

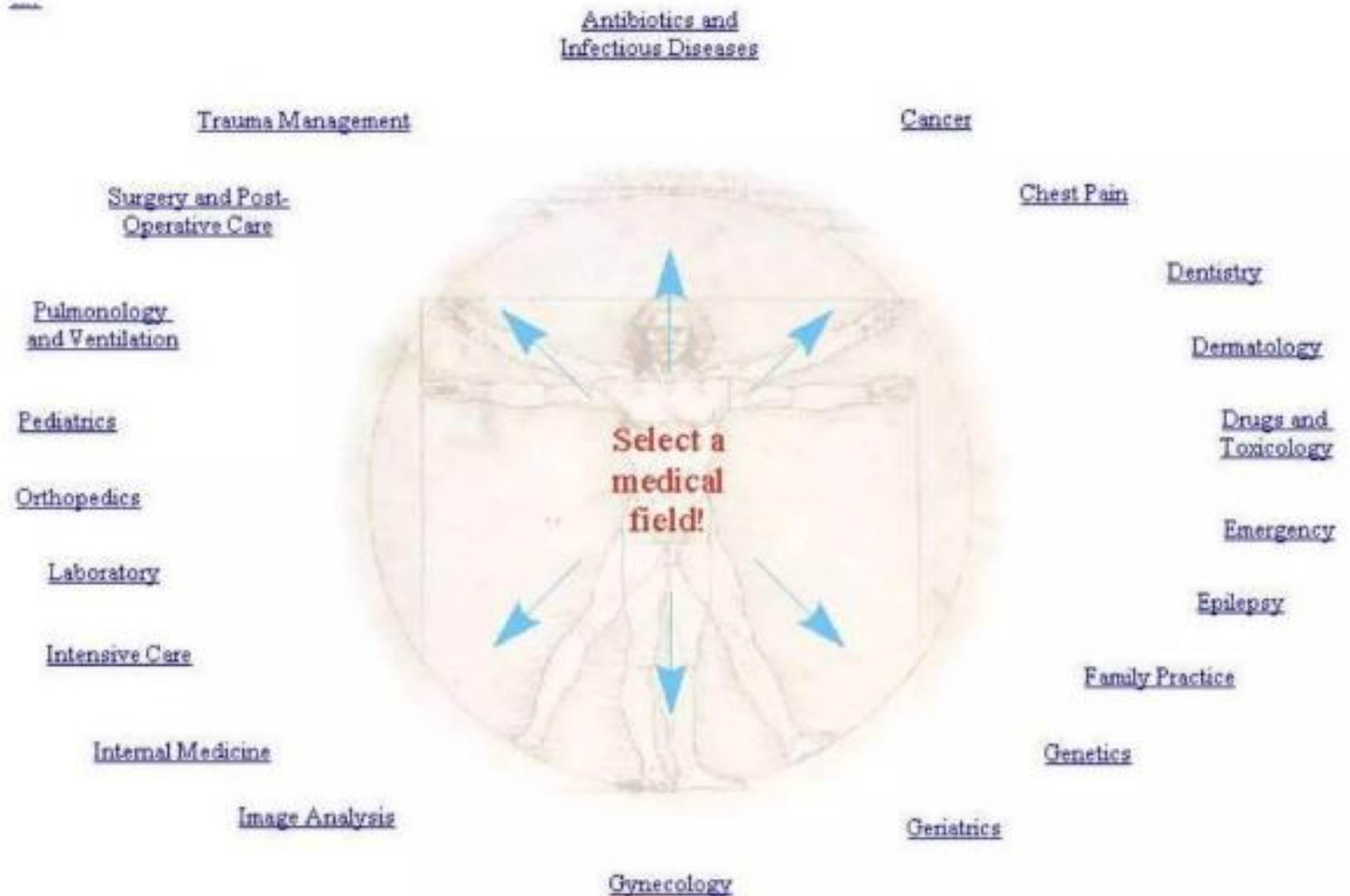
- Biosignals, also known as physiological or biopotential signals, are electrical, mechanical, or chemical signals generated by our body (and living organisms in general, too).
- These signals carry a wealth of information about our health, emotional responses, and physical performance

Why are bio signals important?

- Biosignals contain useful information that can be used to understand the underlying physiological mechanisms of a specific biological event or system, and which may be useful for medical diagnosis.

- Let's see a few examples of information that you can extract from biosignals using biosignal processing methods to demonstrate its applications using [real biosignals sensors](#):
- [Electrocardiography \(ECG\)](#): Extracting heart rate information and its variability under different physical conditions along with the detection of pathological arrhythmia. For instance, a healthy heart has the ability to adapt its heart rate to different physical conditions (e.g. rest vs. physical activity) without showing any signs of pathological issues.
- [Electromyography \(EMG\)](#): Quantifying the amplitude of EMG signals along the time under controlled and repeated workloads to identify muscle fatigue and potential injury risks.
- [Electroencephalography \(EEG\)](#): Identify different brainwave frequencies and their activity to reveal different cognitive states. For instance, active brain regions generate quickly changing signals (i.e. high-frequency activity) while at rest, the signal shows only slow changes (i.e. low-frequency activity).
- [Respiration](#): Extracting respiration rate and identifying breathing patterns while under physical load vs. at rest to differentiate between normal adjustments of respiration to keep the body running with a sufficient oxygen supply and pathological breathing alterations.

Application Areas



Examples of Biomedical Signals

- ⊙ Electroneurogram (ENG)
- ⊙ Electromyogram (EMG)
- ⊙ Electrocardiogram (ECG)
- ⊙ Electroencephalogram (EEG)
- ⊙ Electrogastrogram (EGG)
- ⊙ Electrooculogram (EOG)
- ⊙ Phonocardiogram (PCG)
- ⊙ Vibromyogram (VMG)
- ⊙ Vibroarthrogram (VAG)

ECG (electrocardiograph)

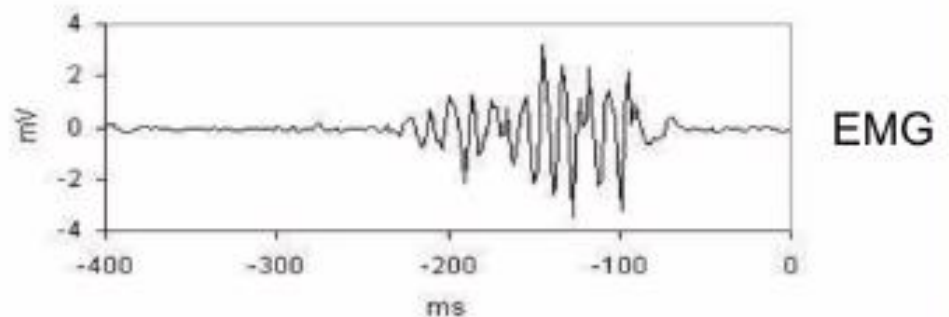


- electrical activity of the heart



EMG (Electromyogram)

signal generated by muscle cells



EGG(Electrogastrogram)

- The electrical activity of the stomach consists of rhythmic waves of depolarization and repolarization of its constituent smooth muscle cells.

EOG(Electroocculogram)

- Eye movements produce a moving (rotating) dipole source and, accordingly, signals that are a measure of the movement are obtained .

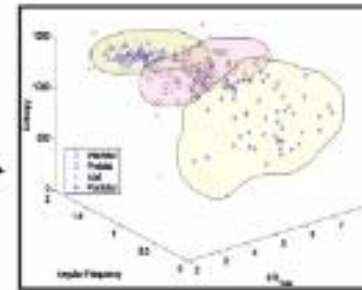


RESEARCH FIELDS

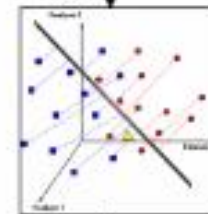
Data Acquisition



Signal Analysis Pre processing



Optimization: Feature Extraction/ Clustering



Statistical Analysis: Pattern Recognition

Initiate a warning or a variety of therapies

Nurse

Stimulator

User/Patient



Drug Technology

1. What are Bio-signals?

Bio-signal is a summarizing term for all kinds of signals that can be measured and monitored from biological beings. Bio-signals are used to understand the underlying physiological mechanisms of a specific biological event or system.

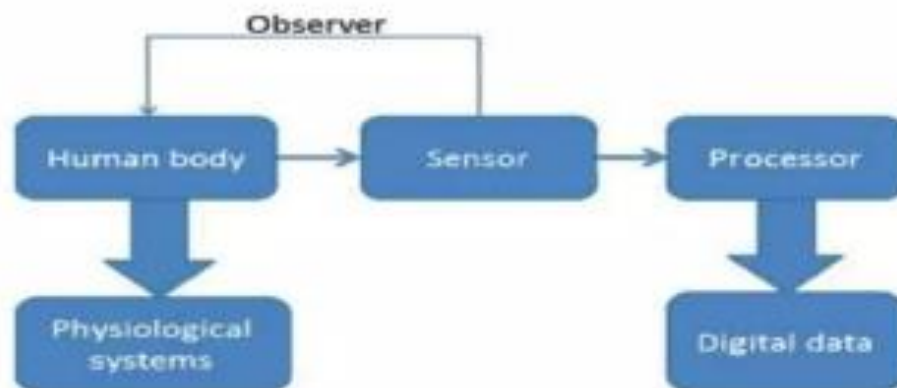
Bio-signals carries all information about the living object. We analyze the signals which are coming from the body (EEG, ECG etc.) or connected to the body (X-Ray images, Ultrasound images).

2. Why bio-signals are important?



Bio-signals are used to understand the underlying physiological mechanisms of a specific biological event or system, and which may be useful for medical diagnosis.

There has been significant research in the medical field on each of these and many more signal types.





3. Objectives of Bio-signals

Biomedical signal processing deals with the innovative applications of signal processing methods in biomedical signals through various creative integrations of the method and biomedical knowledge.

The objectives of Biomedical Signal Processing includes:

- 1- Enhancement of features (waveforms) of interest.
- 2- The quantitative analysis of physiological systems (from cells to organs to the whole human organism).
- 3- To extract useful information from various biological signals and gain a better comprehension of physiological processes or to improve diagnosis, therapy, and rehabilitation in diseased patients.

In general, almost all the signal processing algorithms have the potential to be applied to various biomedical problems.

4. Bio-signals through centuries



Hippocrates palpating a young patient (500 BC)



ICU in modern hospital (2020 AD)

5. Types of Bio-signals

Classification of bio-signals by their generation:

Active Bio-signals:

The energy source for measurement derives from the patient himself (“internal source”).

Examples:

Electrical(bio-potentials): EEG, EMG.

Non-electrical: Temperature, blood pressure.

Passive Bio-signals:

The energy source is from outside the patient (“external source”).

Example: X-ray in CT, Ultrasound.

Sensor

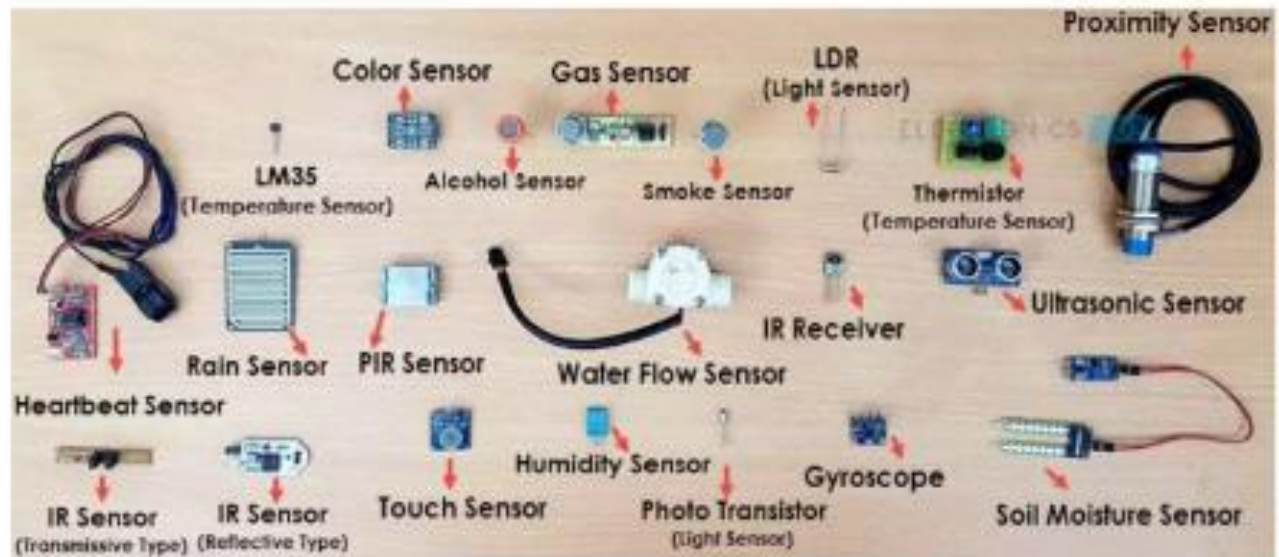
Bio-sensor

Artifacts

Instruments For Measurement

1. Sensors

Sensors are sophisticated devices that are frequently used to detect and respond to electrical or optical signals. A sensor converts the physical parameter (for example: temperature, blood pressure, humidity, speed, etc.) into a signal which can be measured electrically.



2. Bio-sensors



Biosensors are devices used to detect the presence or concentration of a biological analyte, such as a biomolecule, a biological structure or a microorganism. Biosensors consist of three parts: a component that recognizes the analyte and produces a signal, a signal transducer, and a reader device.

Example:

A common example of a commercial biosensor is the blood glucose blood biosensor.



3. Instruments for Measuring Active Bio-signals



Electric Active Bio-signals (Bio-potentials):

Instrument consists of:

- 1- **Electrodes**: enable an electrical conductive connection between the examined body part with the measuring system
- 2- **Signal processor** (amplifier , ADC, electrical filters to remove noise, and unwanted frequencies etc.)
- 3- **Recorder** (also called read-out device, today usually a computer monitor or a chart recorder).

Non-Electric Active Bio-signals:

Electrodes are replaced with appropriate sensors

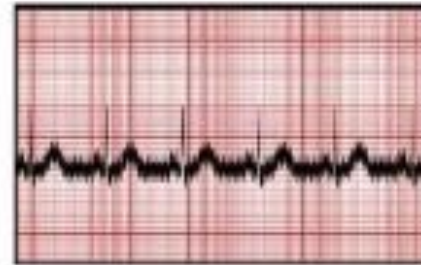
Example: Medical temperature sensors

4.

Bio-signal Artifacts

Bio-signal artifacts may be defined as any unwanted interference which alters the accuracy of the reading obtained from the electrophysiological structure being studied.

Unfortunately a lot of these artifacts tend to have frequencies quite similar to bio-signals desired to be measured and often with a greater amplitude.



50Hz AC superimposed on the EKG



Muscle tremors



Moving baseline from patient movement, dirty electrodes, loose electrodes



Classification of bio signals

1. Classification of Bio-signals

According to Nature of Bio-signals:

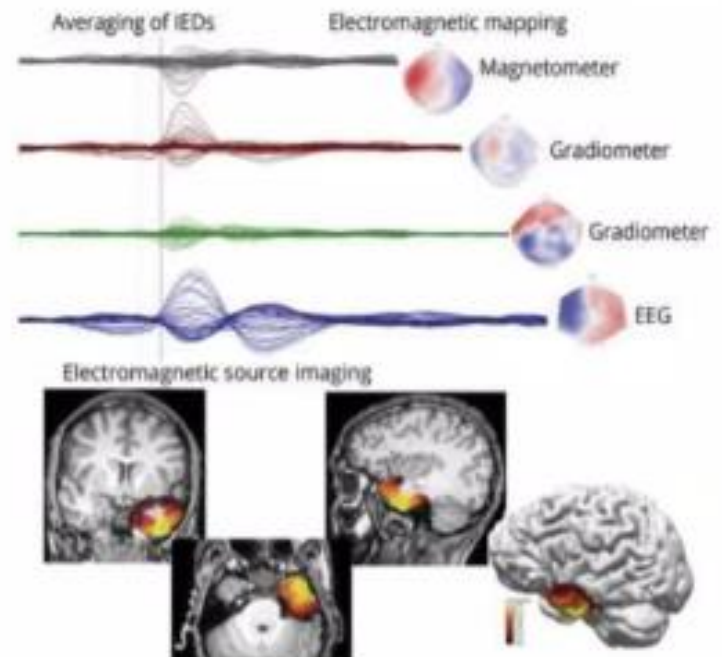
Electric	[example: Electrical activity of heart].
Magnetic	[example: Brain's magnetic field].
Chemical	[example: Level of glucose].
Mechanical (acoustic)	[example: Blood pressure].
Acoustic	[example: Respiratory sounds].
Optical	[example: Blood oxygenation].
Thermal	[example: Heat loss or absorption].

2.

Classification of Bio-signals

According to System of Origin of Bio-signals

- Endocrine System
- Nervous System
- Cardiovascular System
- Vision System
- Auditory System
- Musculoskeletal System
- Respiratory System
- Gastrointestinal System
- Blood System

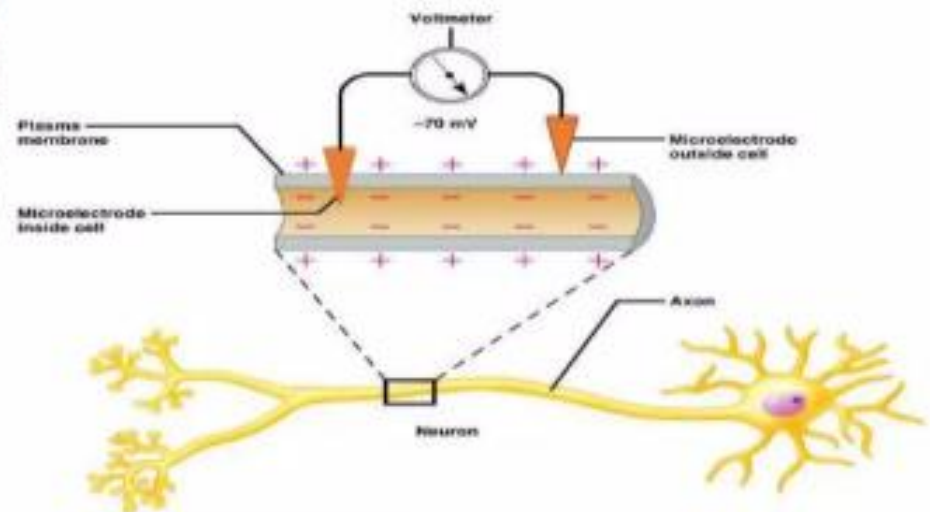
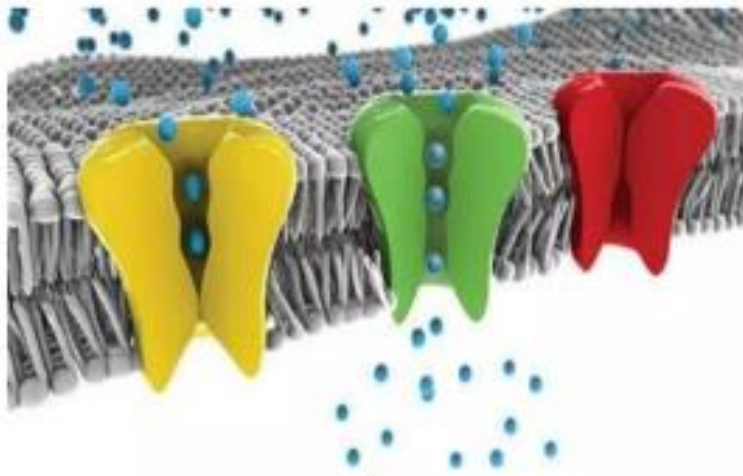


Classification of Bio-signals

According to physical nature of signals and origin

1. Electric Bio-signals

Electric field is generated in cells (nerve and muscle) and organs because of intra- and extracellular ionic currents. They are the results of electrochemical processes in the single ionic channels.



Types of Electrical Signals

Neural Cells

- 1- ENG – Electro-neurogram,
- 2- EEG – Electro-encephalogram,
- 3- ERG – Electro-retinogram.

Muscle Cells

- 4- ECG – Electro-cardiogram,
- 5- EMG – Electro-myogram.

Other cells

- 6- FOG – Electro-oculogram,
- 7- GSR – Galvanic Skin Response.

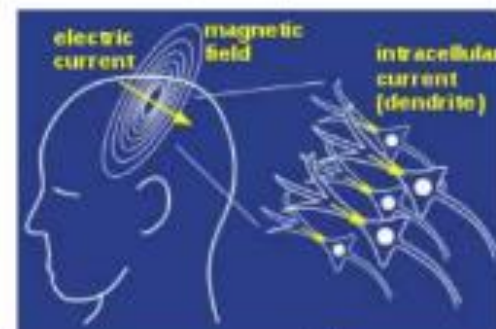


2. Magnetic Bio-signals

Magnetic fields are generated by different organs and cells. These signals include motion and displacement signals, pressure and tension and flow signals, and others.

Magneto-encephalography (MEG)

MEG is a non-invasive neurophysiological technique that measures the magnetic fields generated by neuronal activity of the brain. The spatial distributions of the magnetic fields are analyzed to localize the sources of the activity within the brain, and the locations of the sources are superimposed on anatomical images, such as MRI, to provide information about both the structure and function of the brain



Magnetic Bio-signals

Magnetic fields generated by different organs and cells.

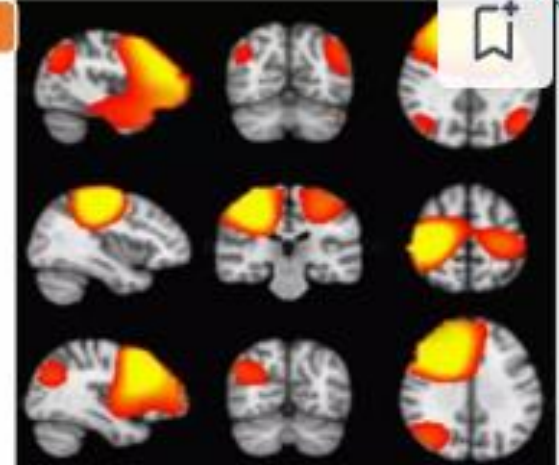
Neural Cells

- 1- MNG – magneto-neurogram
- 2- MEG - magneto-encephalogram.

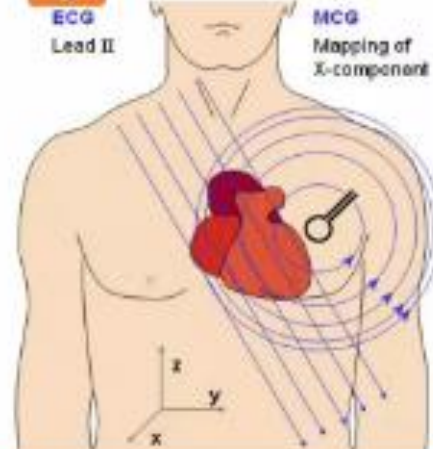
Muscle Cells

- 3- MCG – magneto-cardiogram,
- 4- MMG – magneto-myogram

2



3



3



3. Chemical Bio-signals



Signals providing information about concentration of various chemical agents in the body.

- 1- Level of glucose (diabetes).
- 2- Blood oxygen level (asthma, obstructive pulmonary disease, heart and kidney failure).
- 3- Gases in blood and breathing airflow (anesthetic gases, carbon dioxide etc.)
- 4- pH levels.



4



2



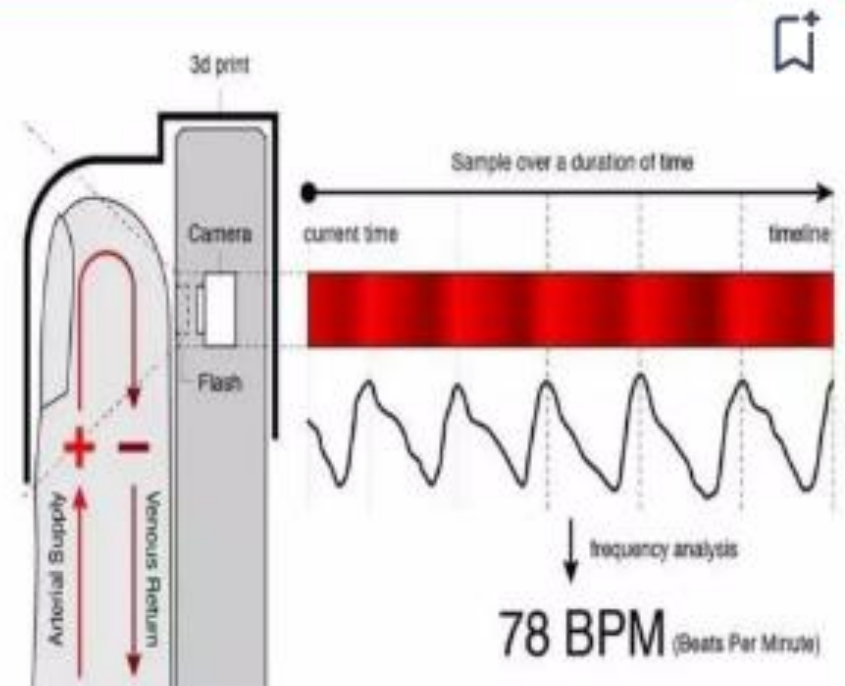
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Chemical Bio-signals

Photoplethysmogram (PPG)

PPG is an optically obtained plethysmogram that can be used to detect blood volume changes in the microvascular bed of tissue.

A **PPG** is often obtained by using a pulse oximeter which illuminates the skin and measures changes in light absorption.



Chemical Bio-signals

Blood Saturation of Gases

SaO₂ - Arterial blood oxygen saturation, describe the percentage of hemoglobin molecules carrying oxygen.

SvO₂ - Venous oxygen saturation, describes how much oxygen the body consumes.

SpO₂ - Peripheral capillary oxygen saturation - the same as SaO₂ but in the capillary system.

SpCO₂ - Concentration of carboxyhemoglobin in blood.

4. Mechanical Bio-signals

Bio-mechanical signals reflect **mechanical functions** of body parts.

Examples:

- 1- Blood pressure.
- 2- Accelerometer signals describing human movement, GAIT.
- 3- Spirometry – lung volumes and capacities.
- 4- Phonocardiogram - heart sounds.
- 5- Apexcardiography – motion of heart muscles, velocity and volume of heart filling.
- 6- Tonometry – intraocular pressure.
- 7- Ergometry – physical work activity.

6

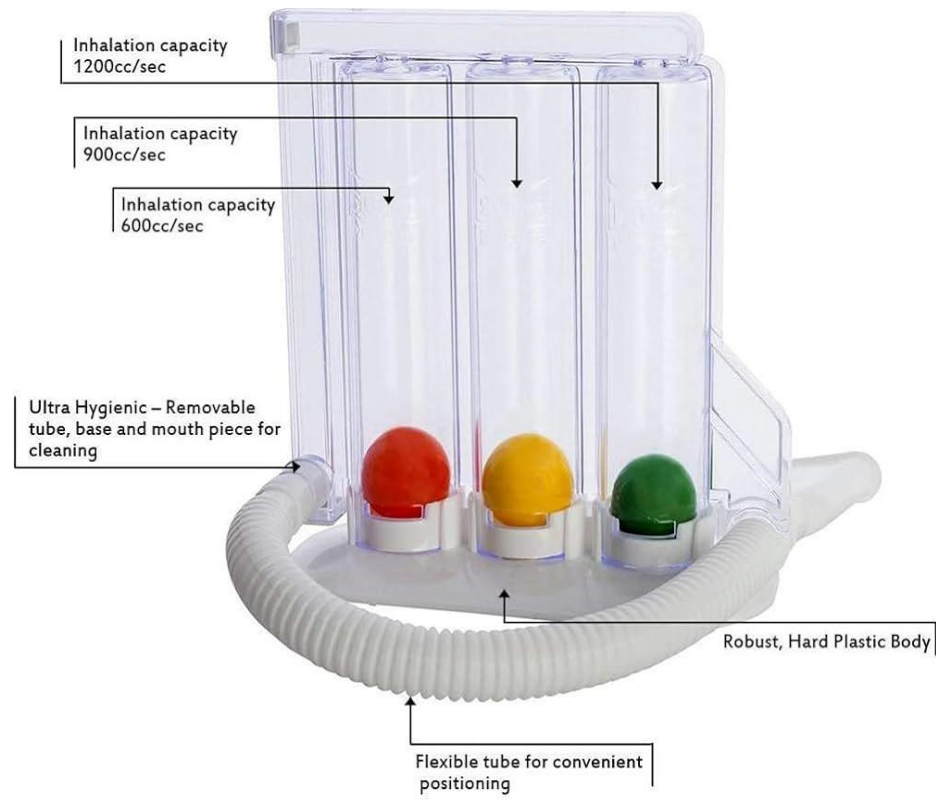
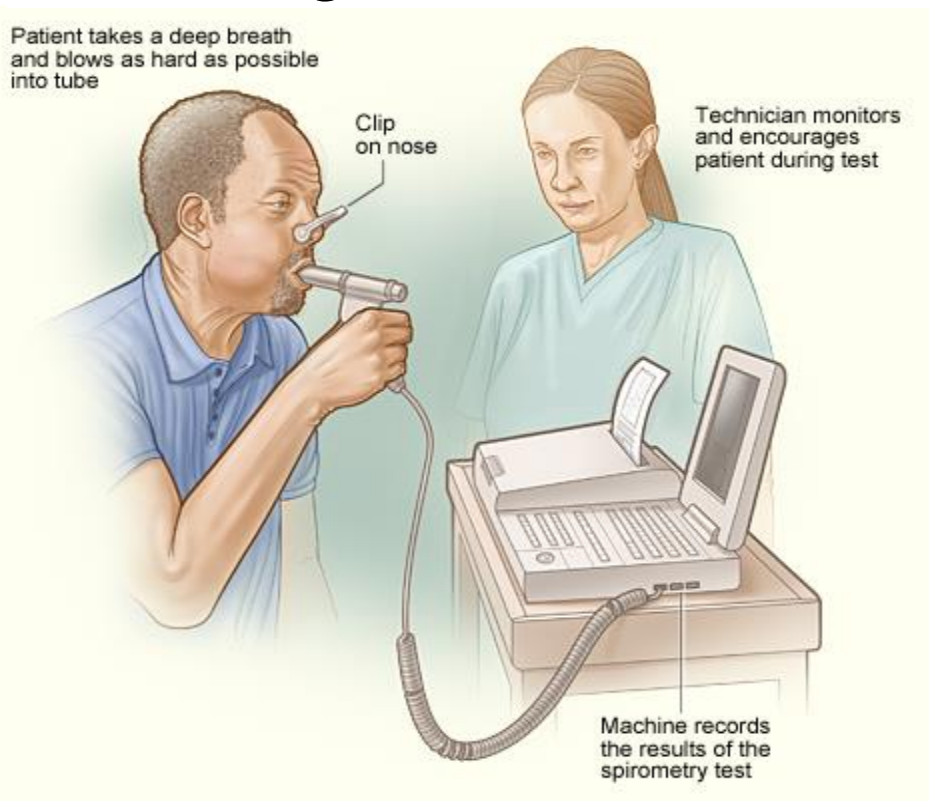


7



Mechanical Bio-Signals

- Spirometry is a common office test used to assess how well your lungs work by measuring how much air you inhale, how much you exhale and how quickly you exhale.
- Spirometry is used to diagnose asthma, chronic obstructive pulmonary disease (COPD) and other conditions that affect breathing.



5. Acoustic Bio-signals

Subset of mechanical signals that describe the acoustic sound produced by the body (vibrations and motions).

Bio-acoustic signals give access to diverse body sounds:

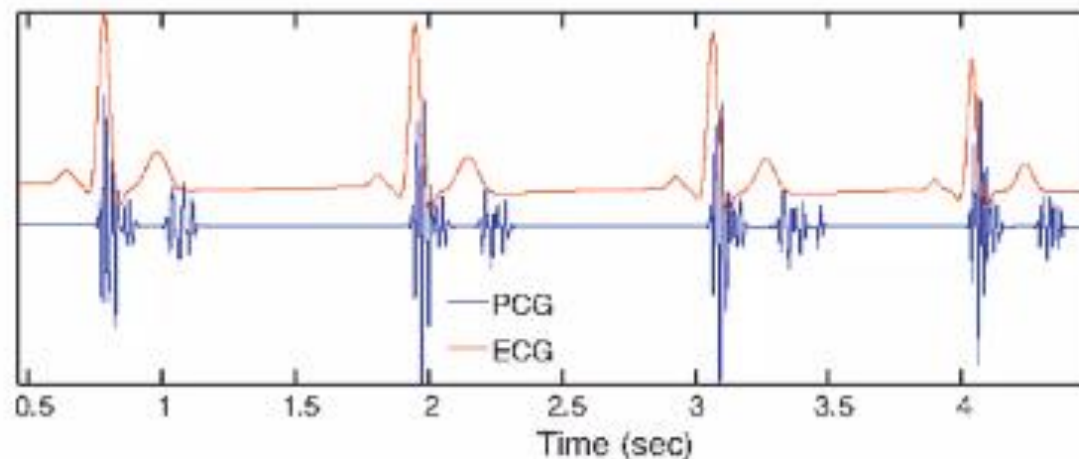
- 1- Cardiac sounds (phonocardiography).
- 2- Snoring (Obstructive Sleep Apnea detection).
- 3- Swallowing.
- 4- Respiratory sounds.
- 5- Crackles of joints and muscles.



Acoustic Bio-signals

Phonocardiogram (PCG)

PCG reflects sounds of heartbeats, produced by heart sounds corresponding to two consecutive heart valve closures. Indicates closure strength and the valve's stiffness.





Acoustic Bio-signals

Respiratory sounds

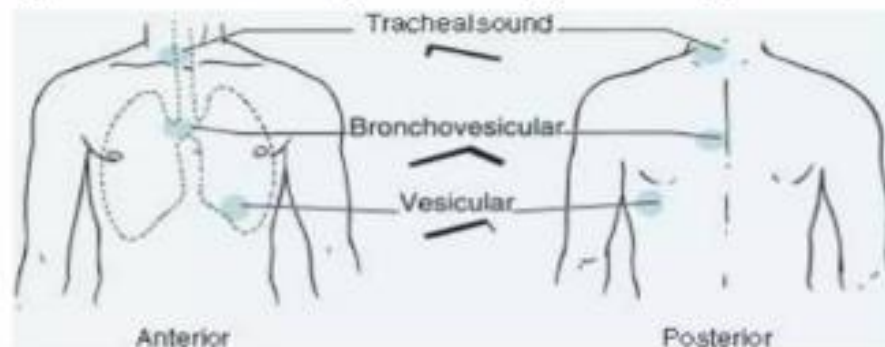
Reflect normal breathing sounds superimposed with crackles, cough sounds, rhonchus, snoring, squawk, stridor and wheeze sounds, which are associated with pulmonary disorders.

Rhonchi (a low-pitched breath sound)

Crackles (a high-pitched breath sound)

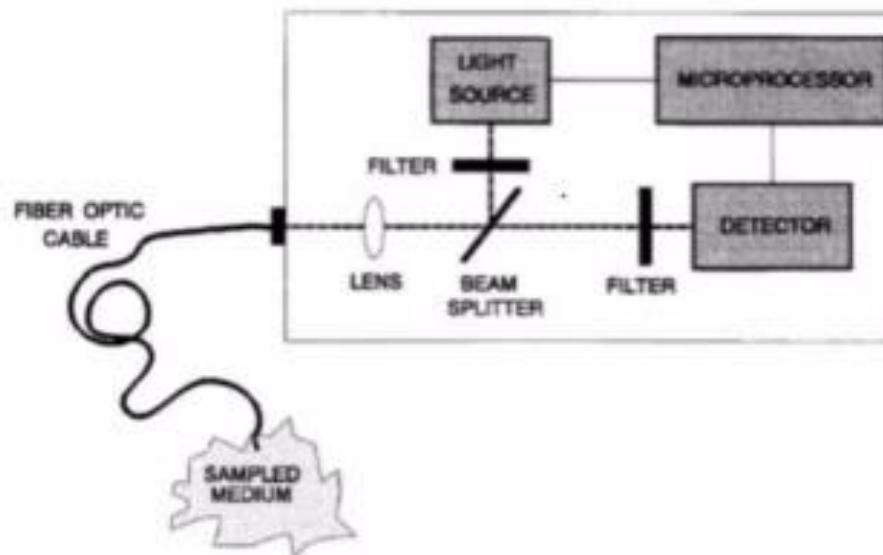
Wheezing (a high-pitched whistling sound caused by narrowing of the bronchial tubes)

Stridor (a harsh, vibratory sound caused by narrowing of the upper airway)



6. Optical Bio-signals

Optical methods are among the oldest and best-established techniques for sensing biochemical analyses.





Optical Bio-signals

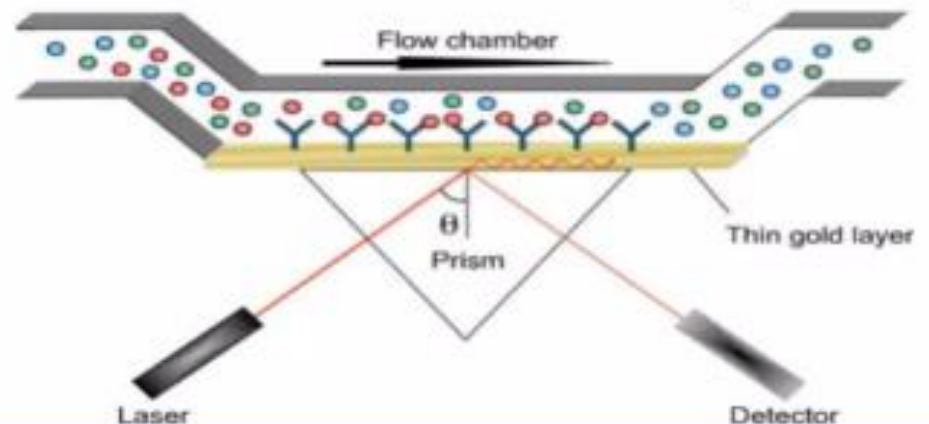
Immunology (anti-gen and anti-body interaction)

Surface Plasmon Resonance (SPR)

SPR is the resonant oscillation of conduction electrons at the interface between negative and positive permittivity material stimulated by incident light. Surface plasmon resonance is the collective oscillation of electrons stimulated by incident light

Example:

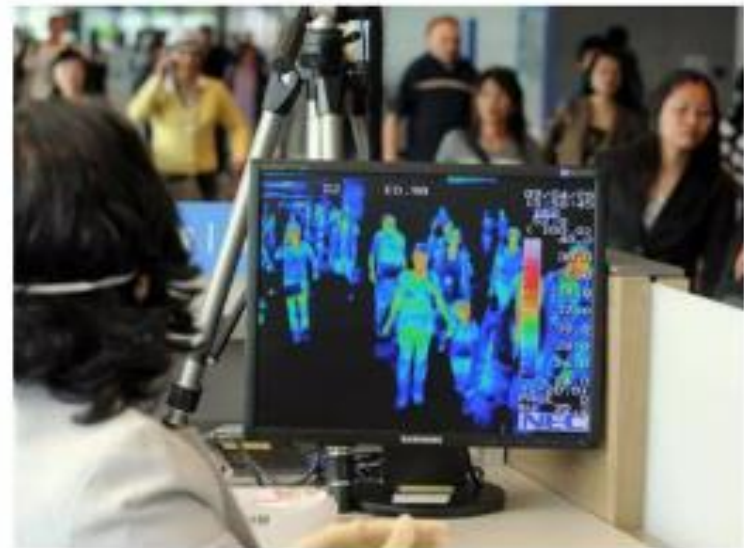
SPR can help to observe, in real-time, biomolecular interactions between a variety of proteins, DNA/RNA and small molecules.



7. Thermal Bio-signals



Body temperature in the point and temperature maps, may describe heat loss and heat absorption in the body, or temperature distribution over the body surface.



Thermal Bio-signals

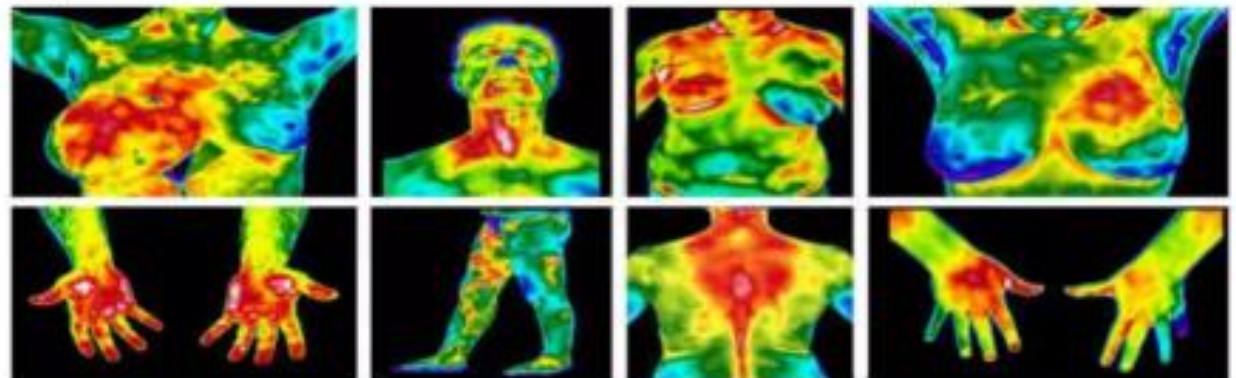


Thermography

It is a test that uses an infrared camera to detect heat patterns and blood flow in body tissues. Digital infrared thermal imaging (DITI) is the type of thermography that's used to diagnose breast cancer.

Example:

Unlike other imaging procedures, thermography doesn't emit any radiation. Breast cancer thermography is a helpful addition to breast cancer screening tests.



THANK YOU